The Importance of Biodiversity

Exam code: 0610 & 0970

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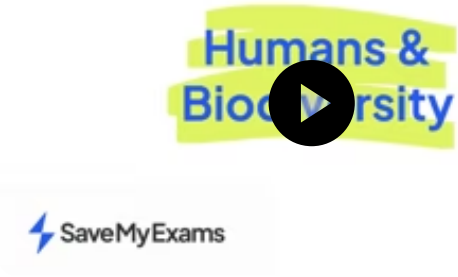
Written by: Phil
Reviewed by: Lára Marie McIvor
Updated on 25 September 2024



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Help me understand

Human Pressures on Other Species

Biodiversity

- Is defined as the **number of different species** that live in a particular area
- Human activities have tended to **force biodiversity downwards**, whereas, high biodiversity is needed for stable ecosystems
- **Habitat destruction** by humans is a major downward pressure on biodiversity

Reasons for Habitat Destruction

- The increasing human population of the planet is causing destruction of many habitats from rainforest to woodland to marine
- Many habitats are destroyed by humans to make space for other economic activities, or by pollution from these activities, and this reduces the biodiversity of these areas
- This interrupts food chains and webs, meaning that more species may die because their prey is gone
- The main reasons for habitat destruction include:

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

AND HOUSING	– AS THERE IS AN INCREASING POPULATION AND DEMAND FOR FOOD, THE AMOUNT OF LAND AVAILABLE FOR THESE THINGS MUST BE INCREASED BY CLEARING HABITATS SUCH AS FORESTS (DEFORESTATION)
EXTRACTION OF NATURAL RESOURCES	– NATURAL RESOURCES SUCH AS WOOD, STONE AND METALS MUST BE GATHERED TO MAKE DIFFERENT PRODUCTS. – THEREFORE MANY TREES ARE CUT DOWN, DESTROYING FOREST HABITATS. IN ADDITION, SOME RESOURCE EXTRACTION TAKES UP A LARGE AMOUNT OF SPACE – FOR EXAMPLE: MINING, WHICH MEANS THAT THE LAND MUST BE CLEARED FIRST
MARINE POLLUTION	– HUMAN ACTIVITIES LEAD TO THE POLLUTION OF MARINE HABITATS – IN MANY PLACES, OIL SPILLS AND OTHER WASTE POLLUTES THE OCEANS, KILLING SEA LIFE – IN ADDITION, EUTROPHICATION CAN OCCUR WHEN FERTILISERS FROM INTENSIVELY FARMED FIELDS ENTERS WATERWAYS – THIS CAUSES A HUGE DECREASE IN BIODIVERSITY IN THESE AREAS AS MOST AQUATIC SPECIES LIVING IN THESE WATERWAYS DIE FROM LACK OF OXYGEN

Deforestation

- Deforestation is the **clearing of trees** (usually on a large scale)
- If trees are replaced by replanting it can be a **sustainable** practise
- Generally the trees are being cleared for the **land to be used in a different way** (for building, grazing for cattle, planting of monocultures such as palm oil plantations etc) and therefore it is not sustainable
- As the amount of the Earth's surface covered by trees decreases, it causes increasingly negative effects on the environment and is a **particularly severe example of habitat destruction**
- Undesirable effects of deforestation include:
  - **Extinction** of species
  - Loss of **soil**
  - **Flooding**
  - Increase of **carbon dioxide** in the atmosphere

EFFECT	CONSEQUENCE
EXTINCTION / LOSS OF BIODIVERSITY	– FOREST HABITATS, ESPECIALLY TROPICAL RAINFORESTS, HAVE A HUGE RANGE OF BIODIVERSITY AND AS HABITAT IS DESTROYED IT CAUSES THE LOSS OF LARGE NUMBERS OF PLANT AND ANIMAL SPECIES – MANY OF THESE SPECIES ARE ONLY FOUND IN THESE AREAS AND THEREFORE WILL BECOME EXTINCT
SOIL EROSION	– TREE ROOTS HELP TO STABILISE THE SOIL, PREVENTING IT FROM BEING ERODED BY RAIN – TREES WILL USUALLY TAKE UP NUTRIENTS AND MINERALS FROM THE SOIL THROUGH THEIR ROOTS – WITHOUT TREES, NUTRIENTS AND MINERALS WILL REMAIN UNUSED IN THE SOIL SO WILL BE WASHED AWAY INTO RIVERS AND LAKES BY RAIN (LEACHING) – THIS LOSS OF SOIL NUTRIENTS IS PERMANENT AND MAKES IT VERY DIFFICULT FOR FOREST TREES TO REGROW, EVEN IF THE LAND IS NOT CULTIVATED WITH CROP PLANTS OR GRASS FOR CATTLE
FLOODING	– WITHOUT TREES THE TOPSOIL WILL BE LOOSE AND UNSTABLE SO WILL BE EASILY WASHED AWAY BY RAIN, INCREASING THE RISK OF FLASH FLOODING AND LANDSLIDES
INCREASED CARBON DIOXIDE IN ATMOSPHERE	– TREES CARRY OUT PHOTOSYNTHESIS DURING WHICH THEY TAKE IN CARBON DIOXIDE AND RELEASE OXYGEN – THE REMOVAL OF SIGNIFICANT NUMBERS OF TREES MEANS LESS CARBON DIOXIDE IS BEING REMOVED FROM THE ATMOSPHERE (AND LESS OXYGEN RELEASED) – WHEN AREAS OF LAND IN FORESTS ARE CLEARED FOR LAND USE, THE TREES ARE OFTEN BURNED AS OPPOSED TO BEING CUT DOWN. THIS RELEASES CARBON DIOXIDE (IT IS AN EXAMPLE OF COMBUSTION), FURTHER INCREASING CARBON DIOXIDE LEVELS IN THE ATMOSPHERE AND CONTRIBUTING TO GLOBAL WARMING

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